



Module 6.1: Security Devices

- ▶ Endpoint Security | Cisco Networking Academy
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Agenda

- ▶ Firewalls
- ▶ Common Security Architectures
- ▶ Firewall Types
- ▶ IDS and IPS
- ▶ Advantages & Disadvantages of IDS/IPS
- ▶ Types of IPS
- ▶ Specialized Security Appliances
- ▶ Summary & Key Takeaways

Introduction to Security Devices

- ▶ Security devices protect networks from attacks.
- ▶ They monitor, filter, and control network traffic.
- ▶ They help prevent unauthorized access and detect threats early.

Firewalls

- ▶ A firewall controls traffic entering and leaving a network.
- ▶ It enforces security rules based on IP addresses, ports, and protocols.
- ▶ Firewalls are the first line of defense in network security.

Common Security Architectures

- ▶ Perimeter security protects the network edge.
- ▶ A DMZ separates public-facing servers from internal systems.
- ▶ Defense-in-depth uses multiple layers of security.

Firewall Type Descriptions

- ▶ Packet-filtering firewalls inspect packet headers only.
- ▶ Stateful firewalls track active connections.
- ▶ Next-generation firewalls inspect applications and content.

Intrusion Detection Systems (IDS)

- ▶ IDS monitors traffic for suspicious behavior.
- ▶ It generates alerts but does not block traffic.
- ▶ IDS is mainly used for visibility and monitoring.

Intrusion Prevention Systems (IPS)

- ▶ IPS blocks malicious traffic automatically.
- ▶ It is placed inline with network traffic.
- ▶ IPS helps stop attacks before damage occurs.

Advantages and Disadvantages of IDS and IPS

- ▶ IDS does not impact traffic flow but cannot stop attacks.
- ▶ IPS provides strong protection but may cause false positives.
- ▶ Both require regular updates and tuning.

Types of IPS

- ▶ Network-based IPS protects entire networks.
- ▶ Host-based IPS protects individual devices.
- ▶ Wireless IPS protects wireless networks.

Specialized Security Appliances

- ▶ UTM devices combine firewall, IDS/IPS, and antivirus.
- ▶ Web security appliances filter harmful web traffic.
- ▶ Email security appliances block spam and phishing.

Summary & Key Takeaways

- ▶ Security devices are essential for network protection.
- ▶ Firewalls and IDS/IPS play critical roles.
- ▶ Layered security provides the strongest defense.